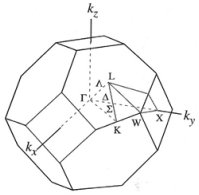
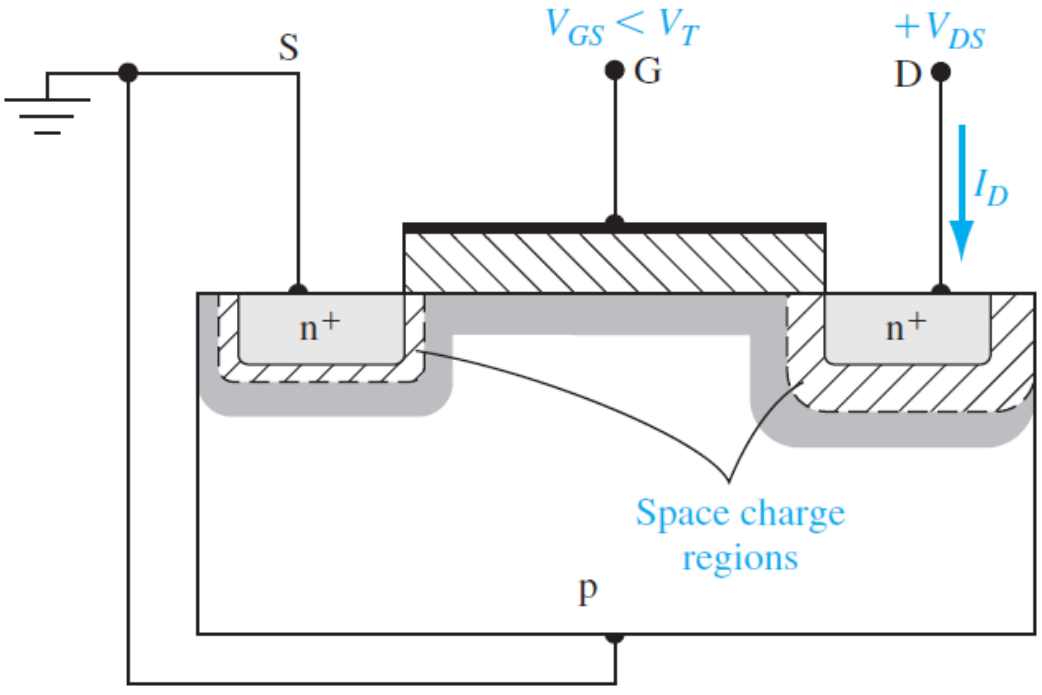
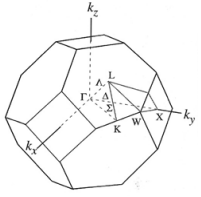
Non-ideal Effects of MOSFET

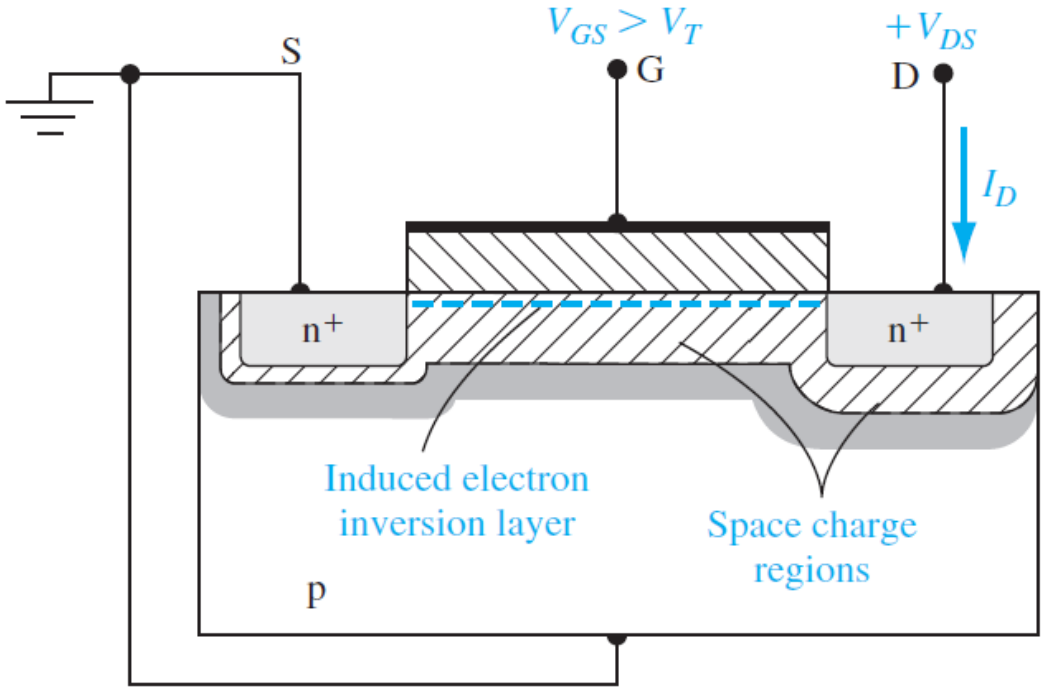


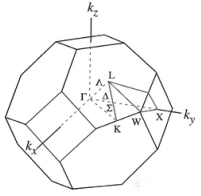
Recall: I-V Relationship $V_{gs}<V_{th}$





Recall: I-V Relationship $V_{gs}>V_{th}$





Subthreshold Conduction

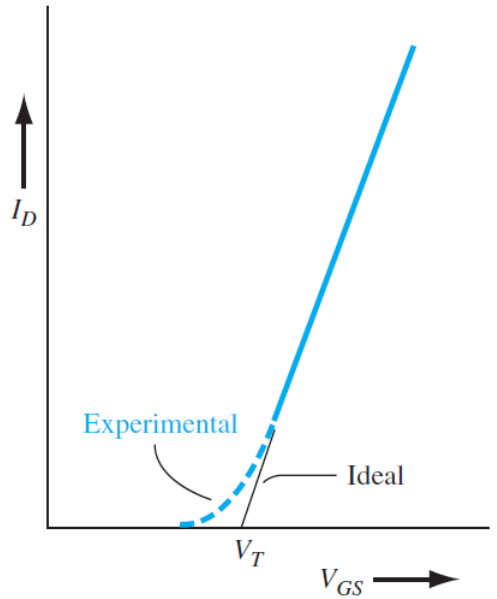


Figure 11.1 | Comparison of ideal and experimental plots of $\sqrt{I_D}$ versus V_{GS}.

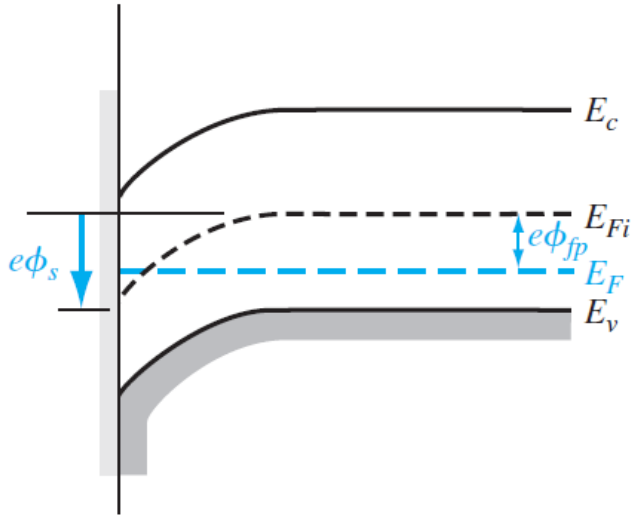
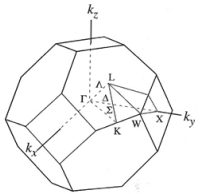


Figure 11.2 | Energy-band diagram when $\phi_{fp} < \phi_s < 2\phi_{fp}$.



Energy-Band

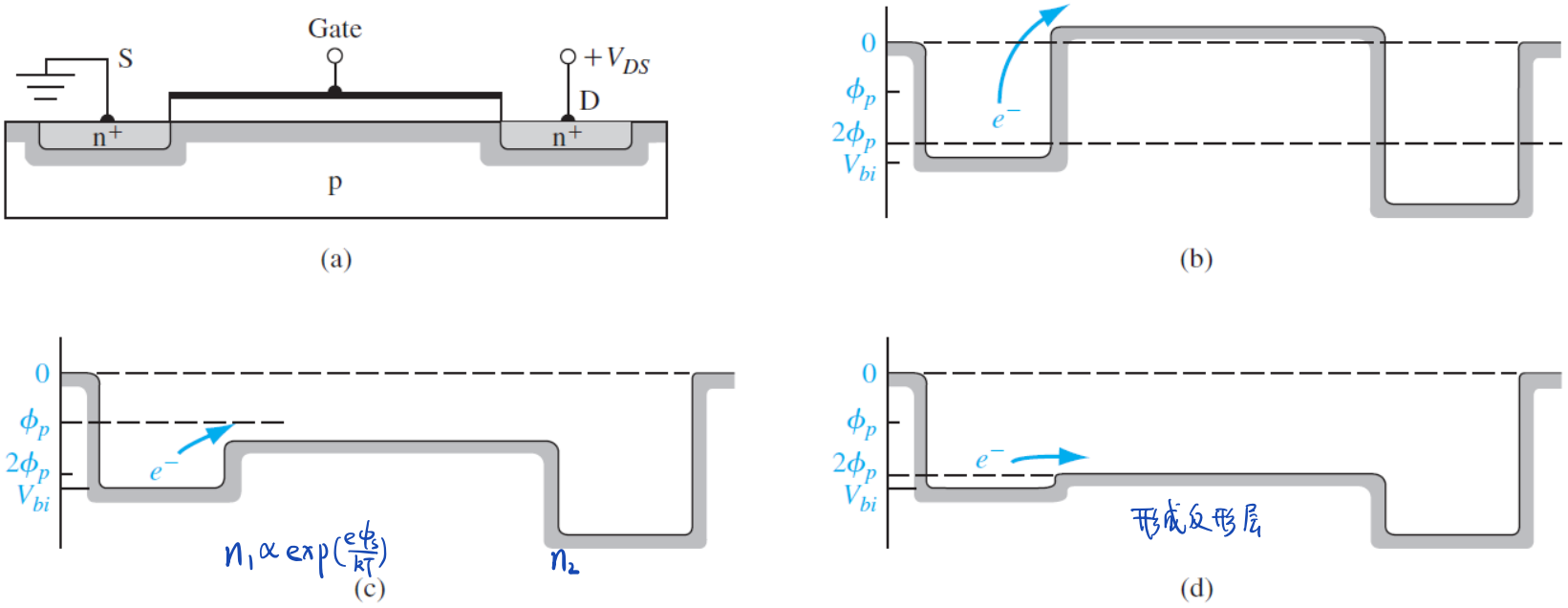
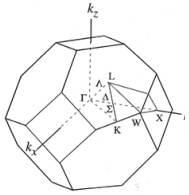


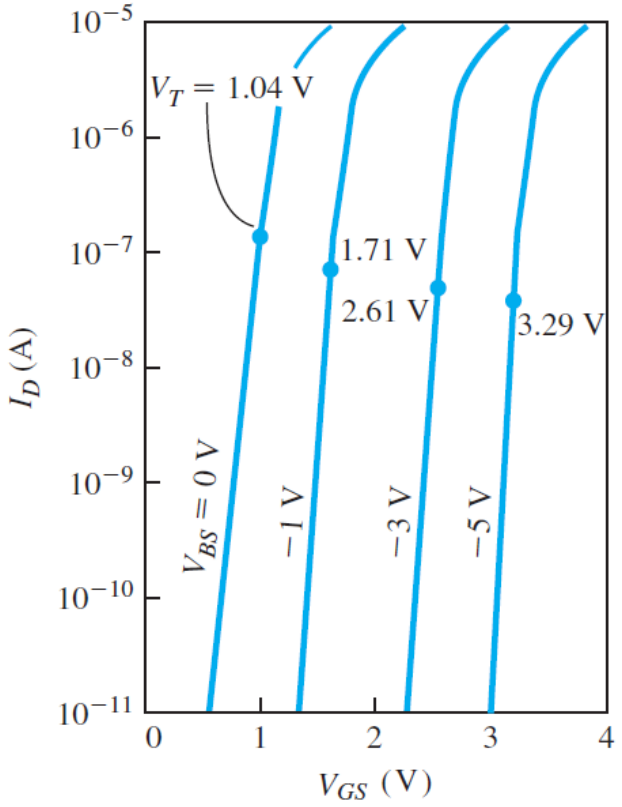
Figure 11.3 | (a) Cross section along channel length of n-channel MOSFET. Energy-band diagrams along channel length at (b) accumulation, (c) weak inversion, and (d) inversion.

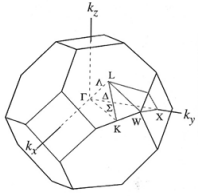


Subthreshold Current

$$I_D(\text{sub}) \propto \left[\exp\left(\frac{eV_{GS}}{kT}\right)\right] \cdot \left[1 - \exp\left(\frac{-eV_{DS}}{kT}\right)\right]$$

The subthreshold current adds significantly to power dissipation in a large-scale integrated circuit.





Channel Length Modulation Effect

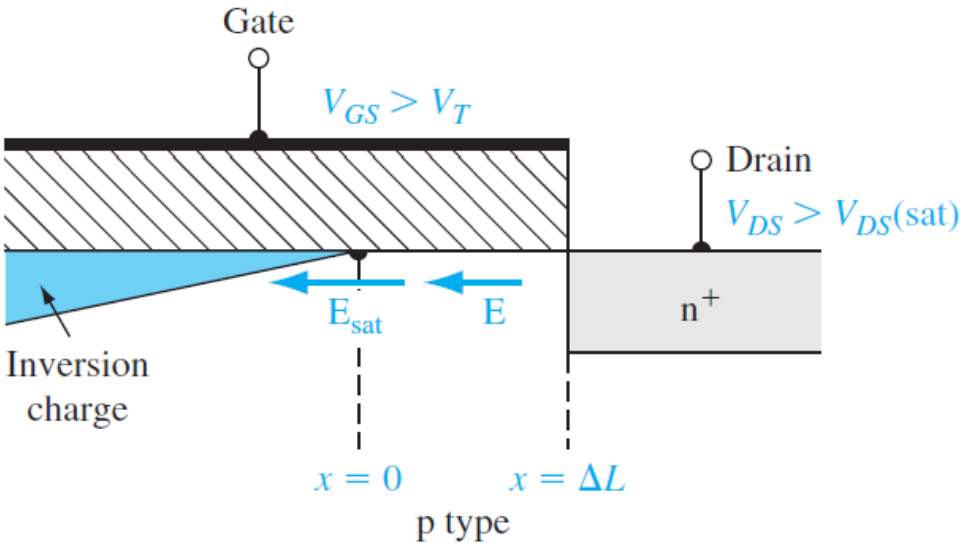
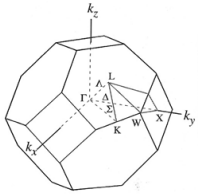


Figure 11.6 | Expanded view of cross section near the drain terminal of an n-channel MOSFET showing the channel length modulation effect.



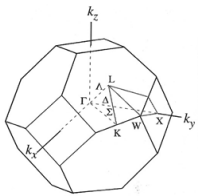
Recall: I-V Relationship Mathematical Derivation

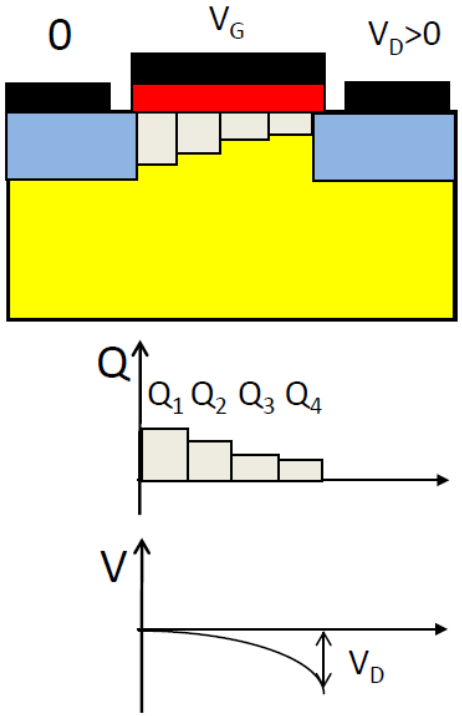
$$|Q_n'| = C_{ox}(V_{GS} - V_x - V_{th})$$

$$I_x = -W\mu_n Q_n' \mathrm{E}_x$$

$$I_x = -W\mu_n C_{\mathrm{ox}} \frac{dV_x}{dx}[(V_{GS} - V_x) - V_T]$$

$$\int_0^L I_x\, dx = -W\mu_n C_{\mathrm{ox}} \int_{Vx(0)}^{Vx(L)} [(V_{GS} - V_T) - V_x]\, dV_x$$





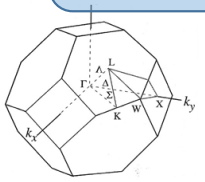
Channel Length Modulation Effect

$$\int_0^L I_x\,dx = -W\mu_n C_{ox}\int_{Vx(0)}^{Vx(L)}[(V_{GS}-V_T)-V_x]\,dV_x$$

$$I_D = \frac{W\mu_n C_{ox}}{2L}[2(V_{GS}-V_T)V_{DS}-V_{DS}^2]$$

$$I'_D = \left(\frac{L}{L-\Delta L}\right)I_D$$

$$I'_D = \frac{k'_n}{2}\cdot\frac{W}{L}\cdot[(V_{GS}-V_T)^2(1+\lambda V_{DS})]$$



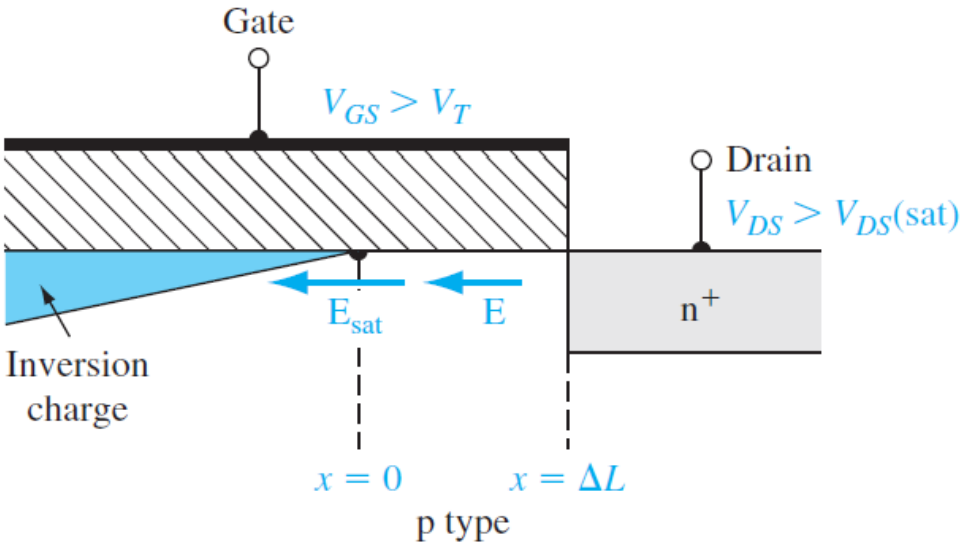


Figure 11.6 | Expanded view of cross section near the drain terminal of an n-channel MOSFET showing the channel length modulation effect.

Channel Length Modulation Effect

$$I'_D = \frac{k'_n}{2} \cdot \frac{W}{L} \cdot [(V_{GS} - V_T)^2 (1 + \lambda V_{DS})]$$

Output Resistance (no longer infinite):

$$r_o = \left(\frac{\partial I'_D}{\partial V_{DS}} \right)^{-1} = \left\{ \frac{k'_n}{2} \cdot \frac{W}{L} \cdot (V_{GS} - V_T)^2 \cdot \lambda \right\}^{-1}$$

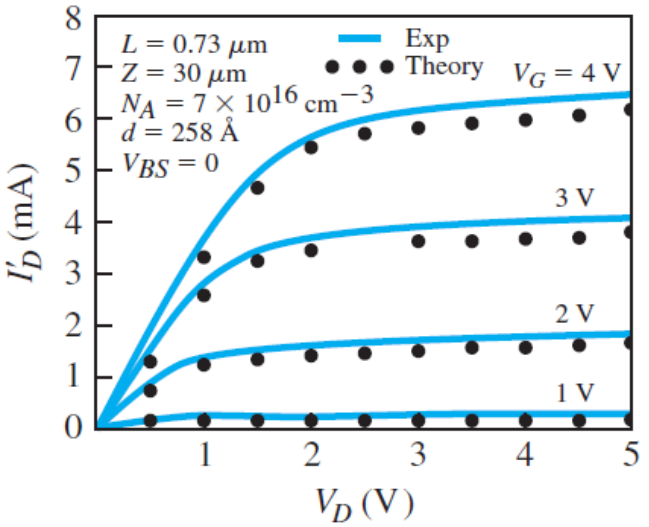
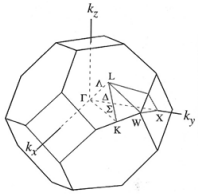


Figure 11.7 | Current–voltage characteristics of a MOSFET showing short-channel effects.
(From Sze [22].)



Mobility Variation

- Surface Scattering

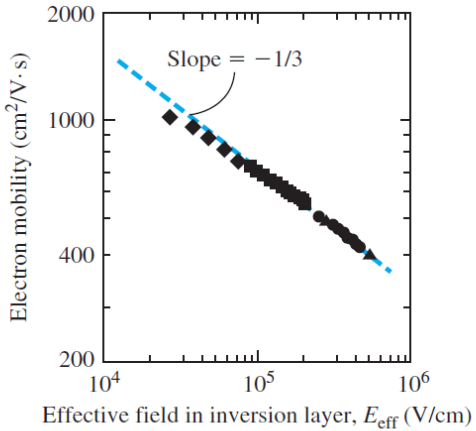


Figure 11.10 | Measured inversion layer electron mobility versus electric field at the inversion layer.
(From Yang [25].)

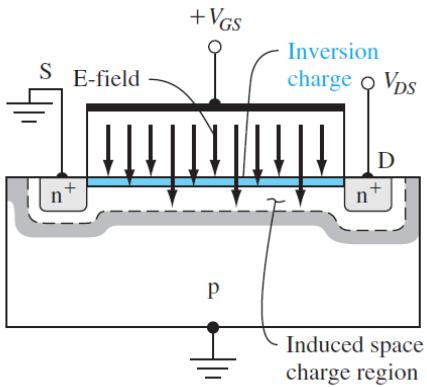


Figure 11.8 | Vertical electric field in an n-channel MOSFET.

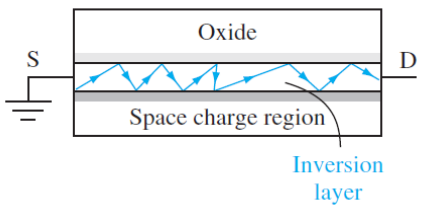
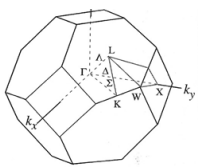
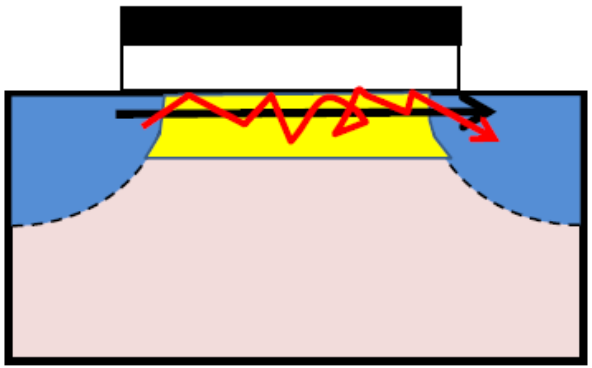
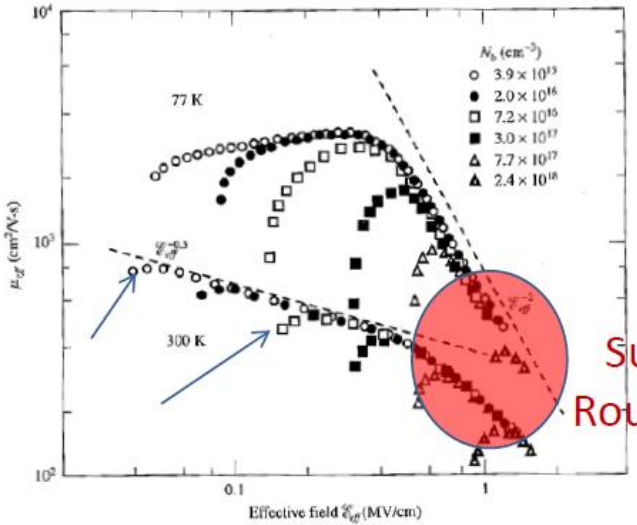


Figure 11.9 | Schematic of carrier surface scattering effects.

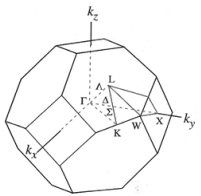


Mobility Enhancement

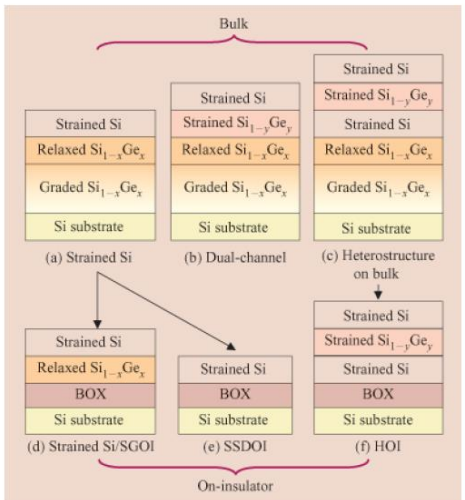


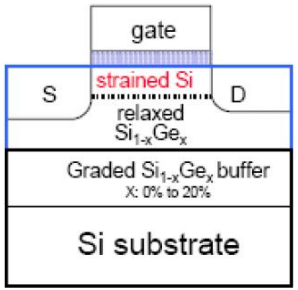


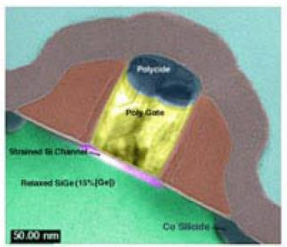
$$I_D = \frac{\mu C_{ox}}{L_{ch}}(V_G - V_{th}^{*})^2$$

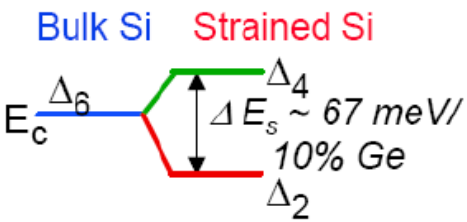


Biaxial Strain to Enhance Mobility

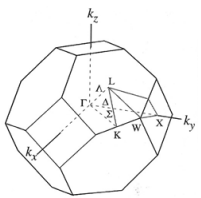




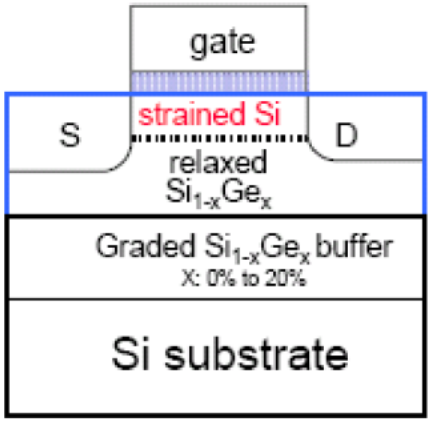


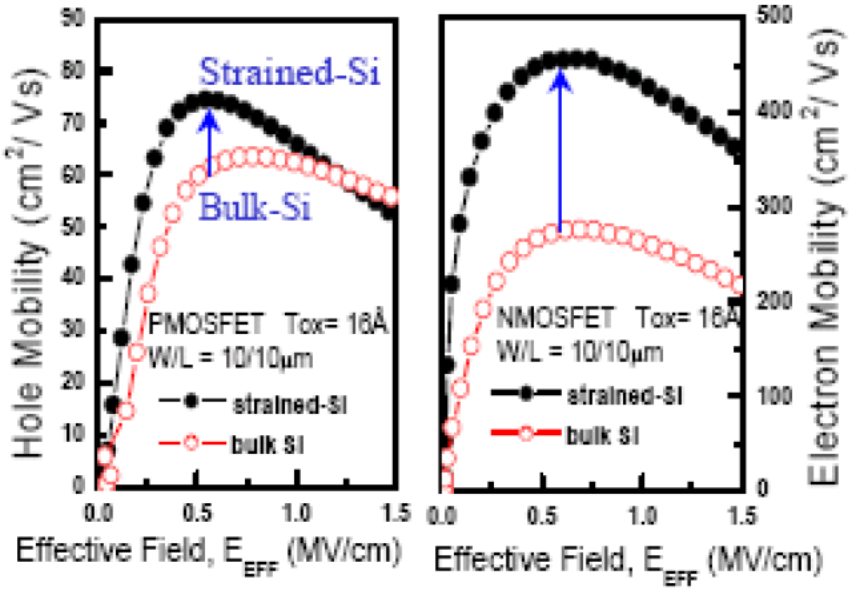


$$\mu = q\frac{\tau}{m_c^*}$$

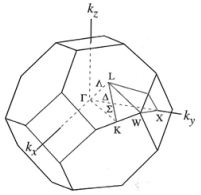


Biaxial Strain to Enhance Mobility

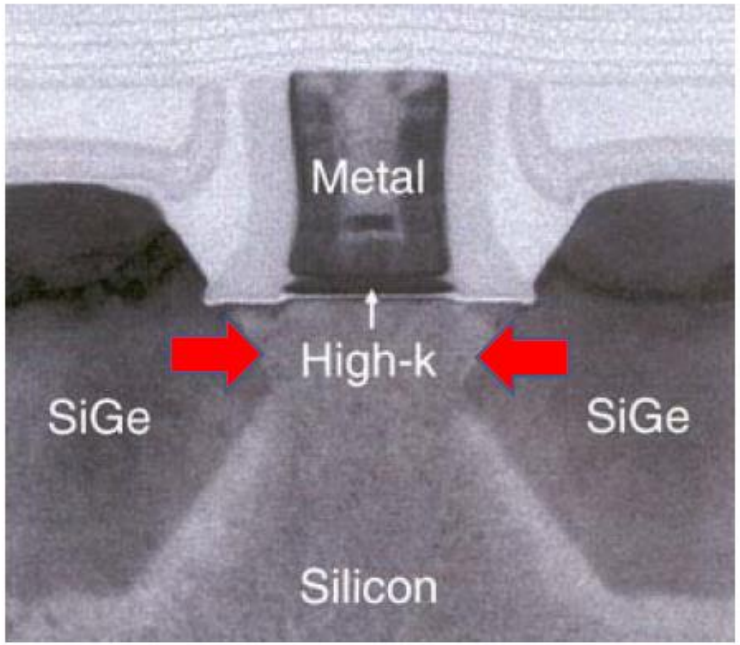


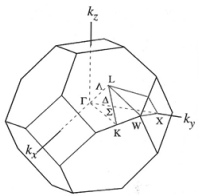


Adapted from Chang et. al, IEDM 2005.

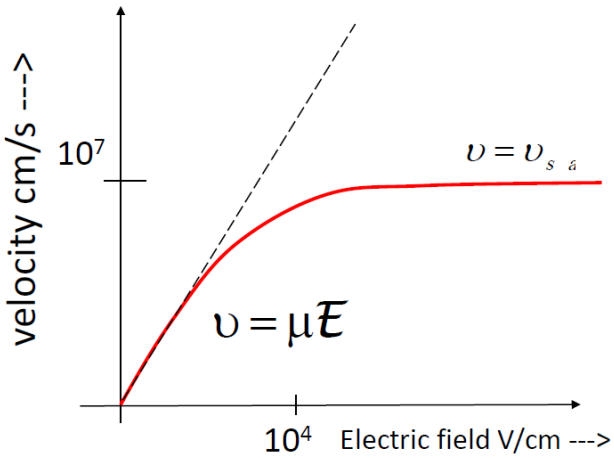


Uniaxial Compressive Strain to Enhance Mobility





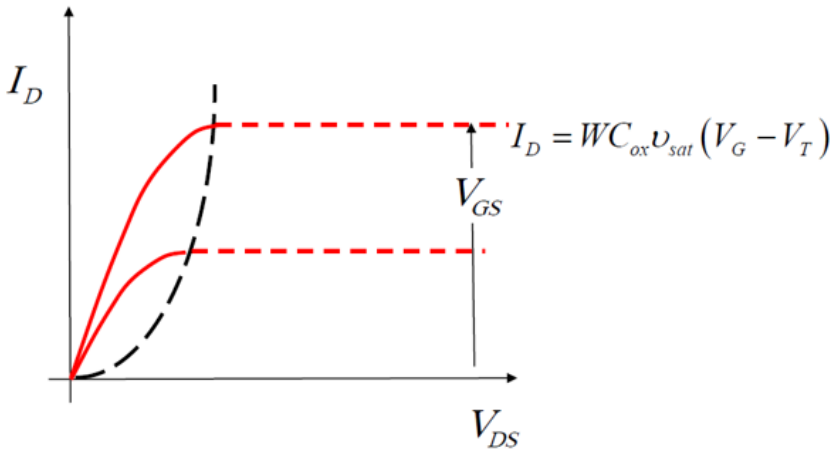
Recall: Velocity Saturation

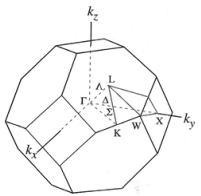


$$\upsilon_d = \frac{-\mu\mathcal{E}}{\left[1+(\mathcal{E}/\mathcal{E}_c)^2\right]^{1/2}}$$

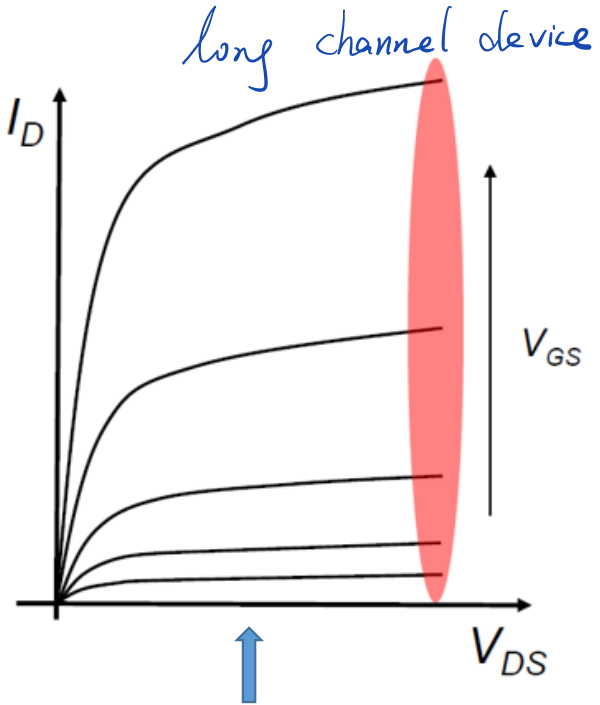
$$\upsilon_d = \frac{-\mu\mathcal{E}}{1+(|\mathcal{E}|/\mathcal{E}_c)}$$

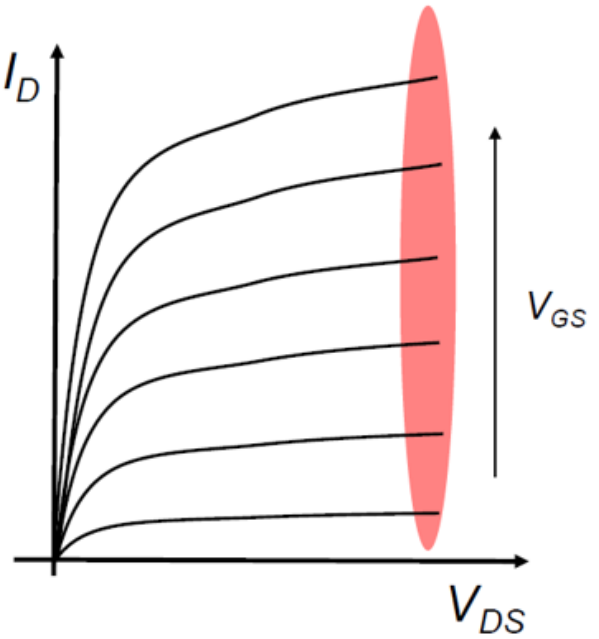
$$\upsilon_{d,sat} = \mu\mathcal{E}_c$$



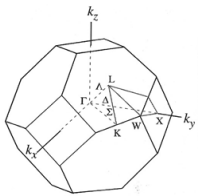


Recall: 'Signature' of Velocity Saturation

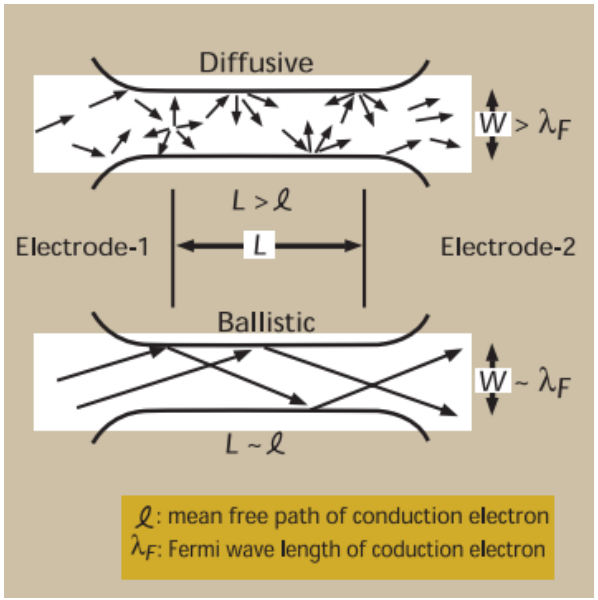


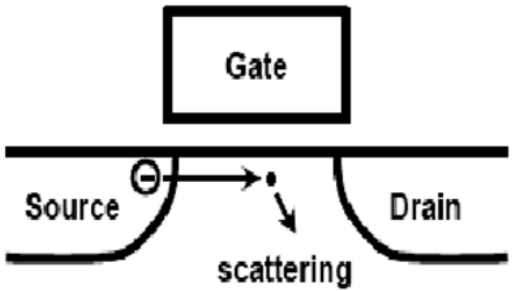


We never see this practically for more than 20 years!

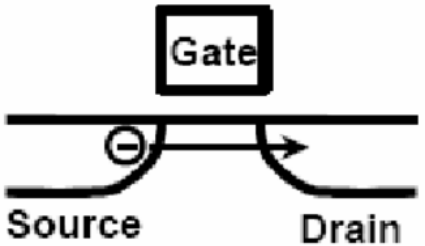


Ballistic Transport (No Scattering)

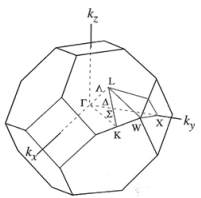




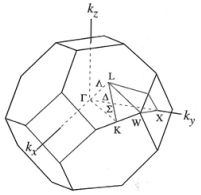
i) Channel Length > Mean Free Path



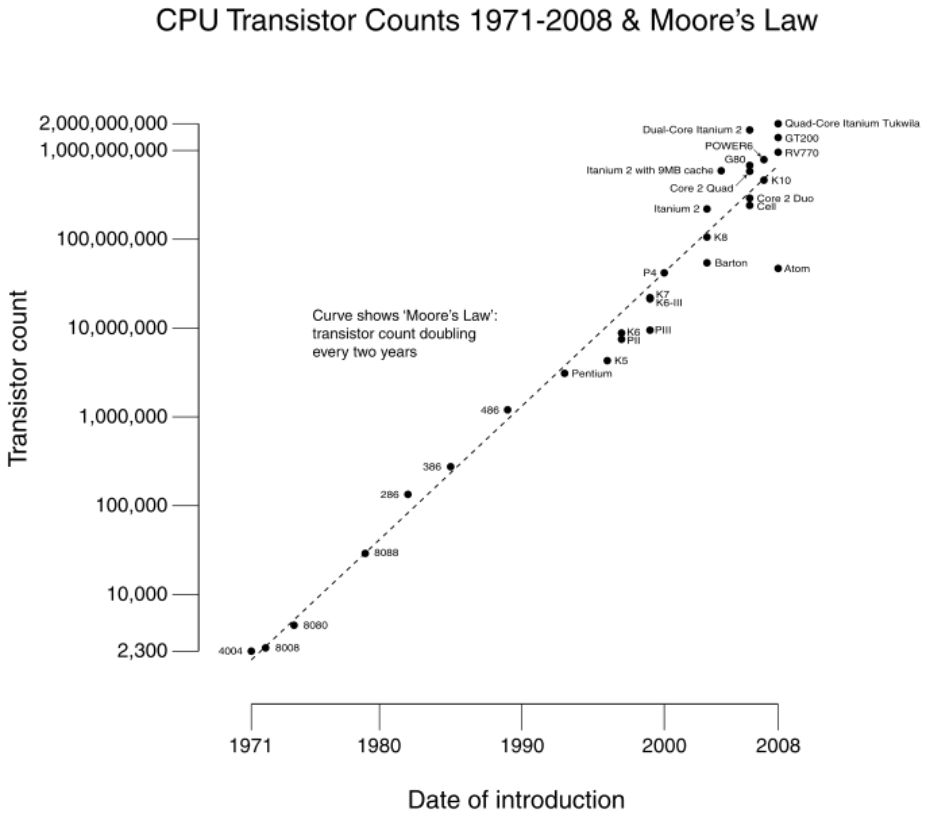
ii) Channel Length < Mean Free Path

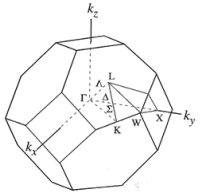


MOSFET Scaling

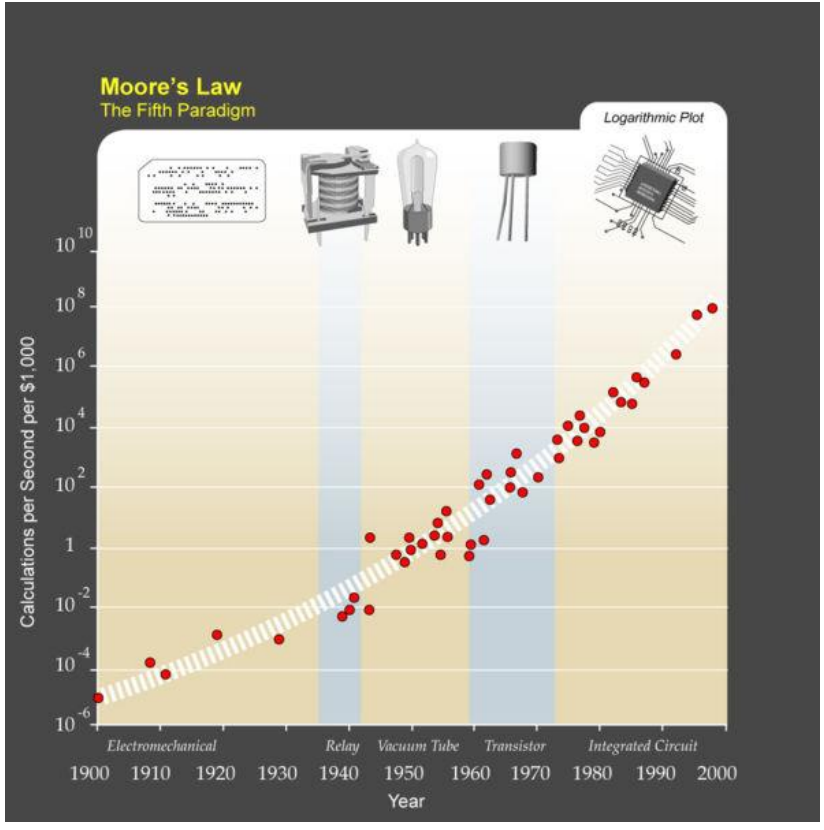


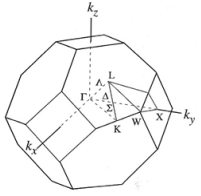
MOSFET Scaling





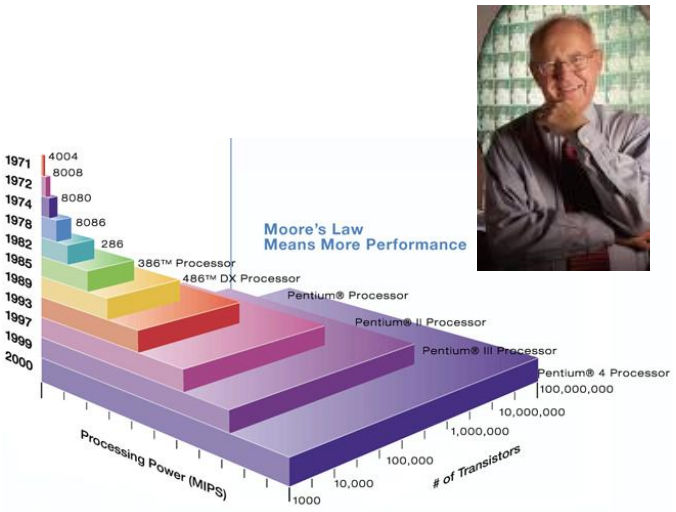
Scaling of Switches

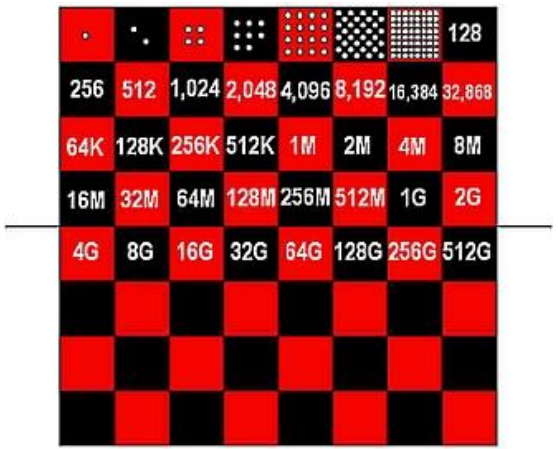




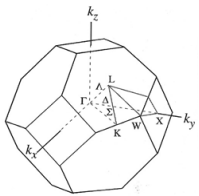
Moore's Law Economics…

- Moore's Law - #DRAM Bits per chip doubles every 18 months
  - ➢~25% bigger chips/wafers
  - ➢~25% design improvements
  - ➢~50 % Lithography – ability to print smaller features

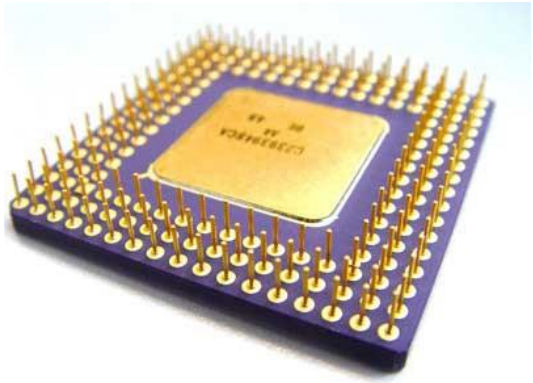




Exponentials !!
(461 B tonnes)



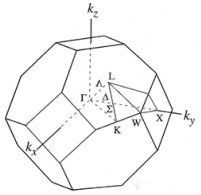
When the Chip's Down…



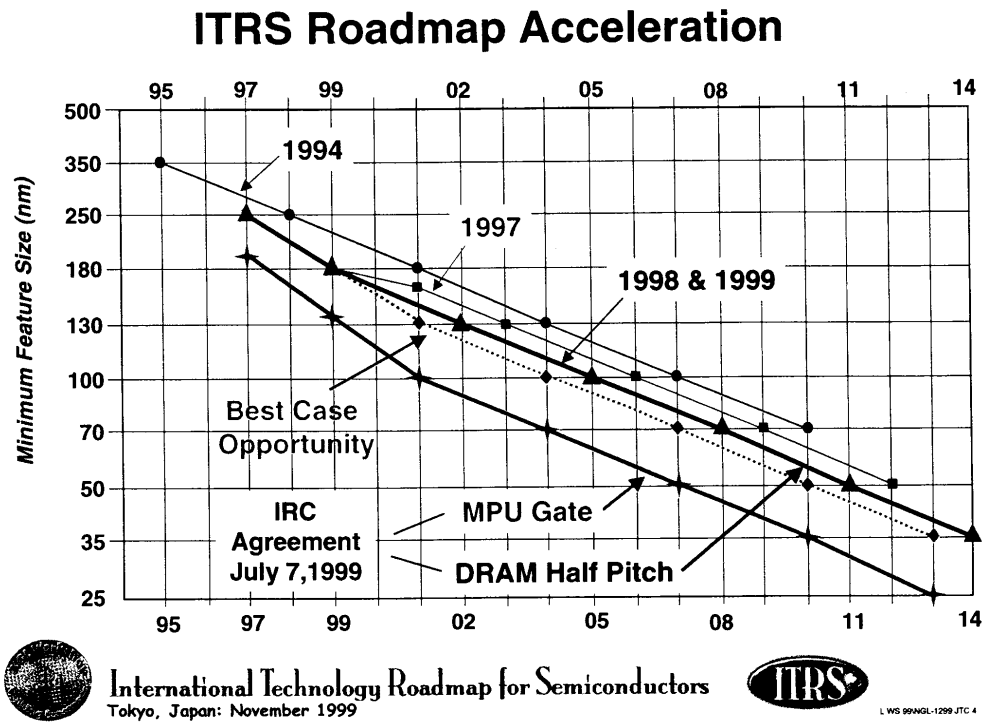
- With feature size shrink of $\sqrt{2}$ (typical generation)
  - 2x #transistors/unit area
  - 2X Higher speed (f_{max})
  - Fixed cost per wafer

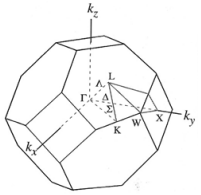
  $\Rightarrow$ Smaller (2x), Faster (2x), cheaper – strong economic driving force

  $\Rightarrow$ 30% improvement in cost per function per year



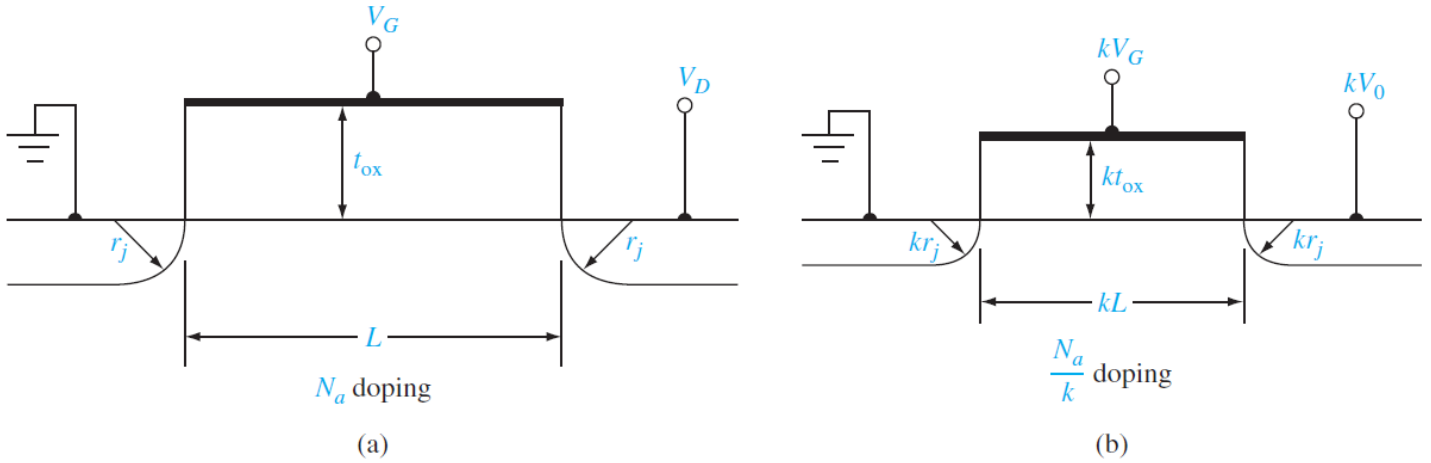
ITRS Roadmap Acceleration

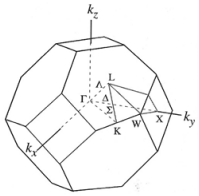




MOSFET Scaling

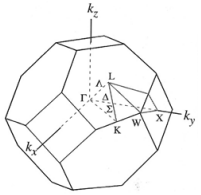
- Constant field scaling





Summary of Constant Field Device Scaling

	Device and circuit parameters	Scaling factor ($k < 1$)
Scaled parameters	Device dimensions (L, t_{ox}, W, x_j)	k
	Doping concentration (N_a, N_d)	$1/k$
	Voltages	k
Effect on device parameters	Electric field	1
	Carrier velocity	1
	Depletion widths	k
	Capacitance ($C = \epsilon A/t$)	k
	Drift current	k
Effect on circuit parameters	Device density	$1/k^2$
	Power density	1
	Power dissipation per device ($P = IV$)	k^2
	Circuit delay time ($\approx CV/I$)	k
	Power−delay product ($P\tau$)	k^3

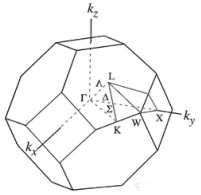


Threshold Voltage

$$V_T = V_{FB} + 2\phi_{fp} + \frac{\sqrt{2\epsilon e N_a (2\phi_{fp})}}{C_{ox}}$$

Depends of the material parameters, and very slight functions of doping

Approximately $\propto \sqrt{k}$



Short Channel Effect

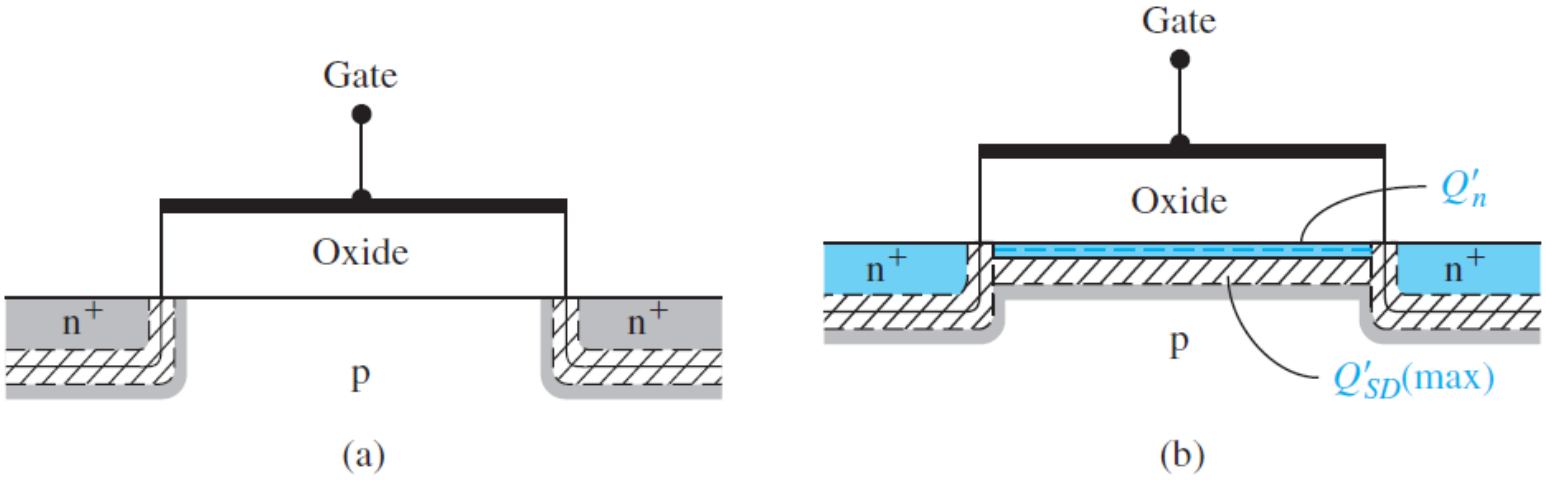
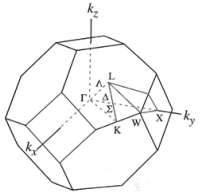


Figure 11.13 | Cross section of a long n-channel MOSFET (a) at flat band and (b) at inversion.

$$V_{TN} = \left(|Q'_{SD}(\max)| - Q'_{ss}\right)\left(\frac{t_{ox}}{\epsilon_{ox}}\right) + \phi_{ms} + 2\phi_{fp}$$



Short Channel Effect

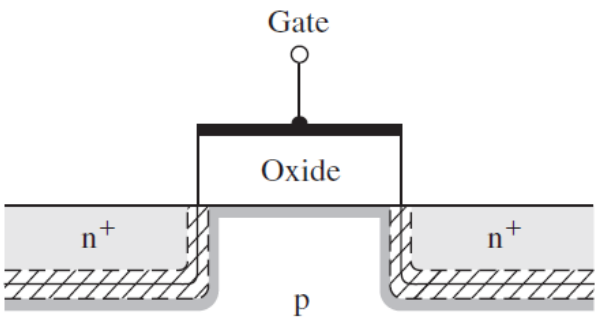
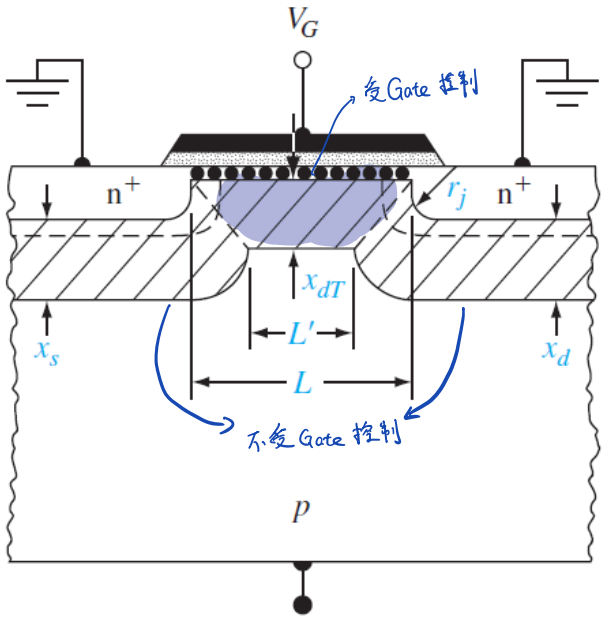
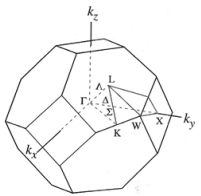


Figure 11.14 | Cross section of a short n-channel MOSFET at flat band.

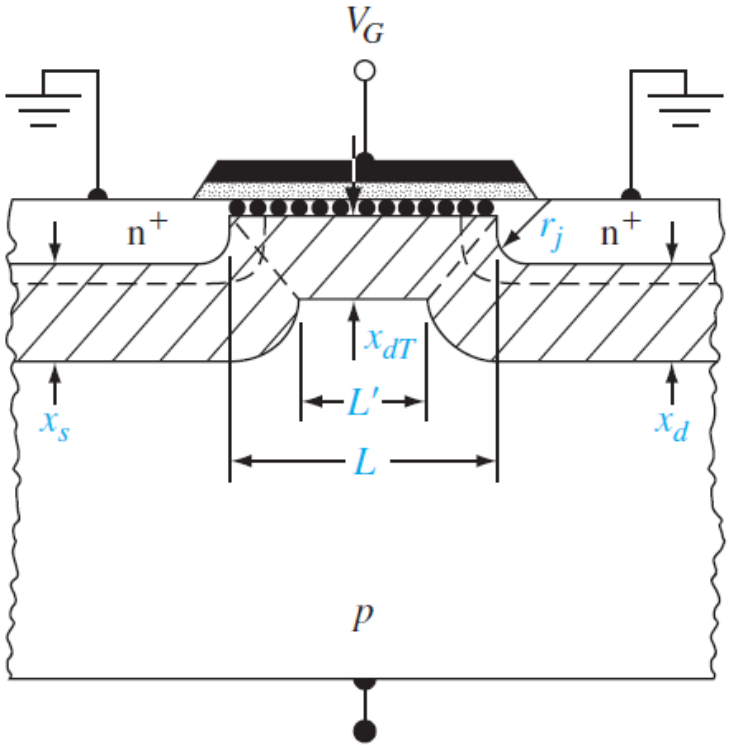


受Gate控制的电荷↓
阈值电压↓

Figure 11.15 | Charge sharing in the short-channel threshold voltage model. *(From Yau [26].)*



Short Channel Effect

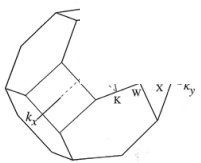


$$|Q'_B| \cdot L = eN_a x_{dT}\left(\frac{L+L'}{2}\right)$$

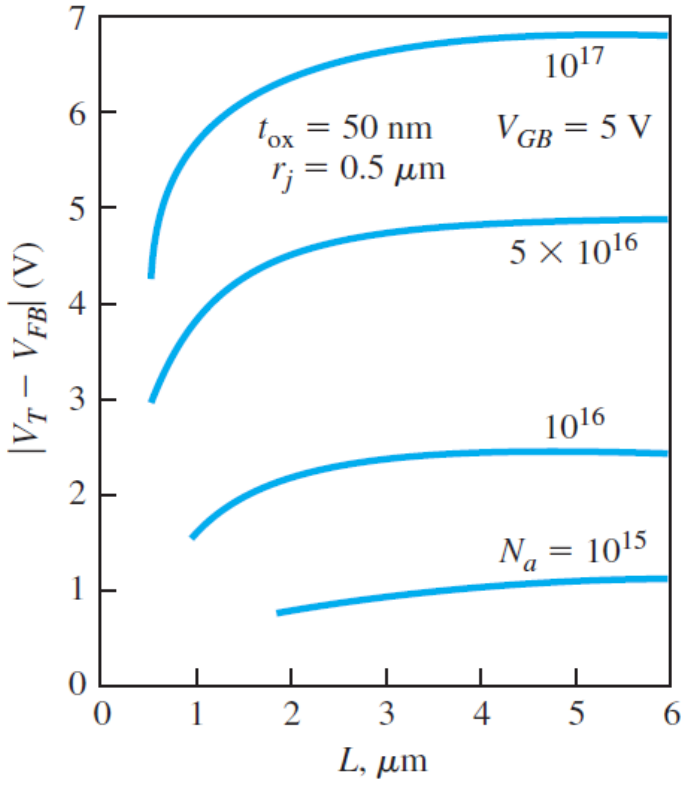
$$\frac{L+L'}{2L} = \left[1 - \frac{r_j}{L}\left(\sqrt{1+\frac{2x_{dT}}{r_j}} - 1\right)\right]$$

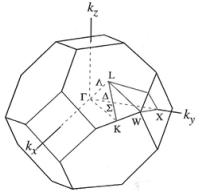
$$|Q'_B| = eN_a x_{dT}\left[1 - \frac{r_j}{L}\left(\sqrt{1+\frac{2x_{dT}}{r_j}} - 1\right)\right]$$

$$\Delta V_T = -\frac{eN_a x_{dT}}{C_{ox}}\left[\frac{r_j}{L}\left(\sqrt{1+\frac{2x_{dT}}{r_j}} - 1\right)\right]$$



Threshold Voltage Versus Channel Length





Narrow Gate Effect

$$Q_B = Q_{B0} + \Delta Q_B$$

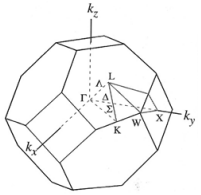
$$|Q_{B0}| = eN_a WLx_{dT}$$

$$\Delta Q_B = eN_a Lx_{dT}(\xi x_{dT})$$

$$|Q_B| = |Q_{B0}| + |\Delta Q_B| = eN_a\, WL\, x_{dT} + eN_a\, Lx_{dT}\,(\xi x_{dT})$$

$$= eN_a WLx_{dT}\left(1 + \frac{\xi x_{dT}}{W}\right)$$

$$\Delta V_T = \frac{eN_a x_{dT}}{C_{ox}}\left(\frac{\xi x_{dT}}{W}\right)$$



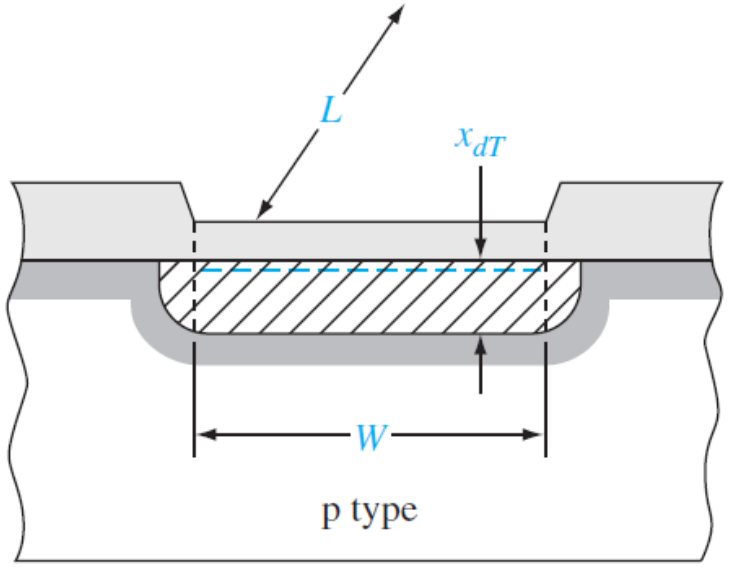


Figure 11.18 | Cross section of an n-channel MOSFET showing the depletion region along the width of the device.

Summary

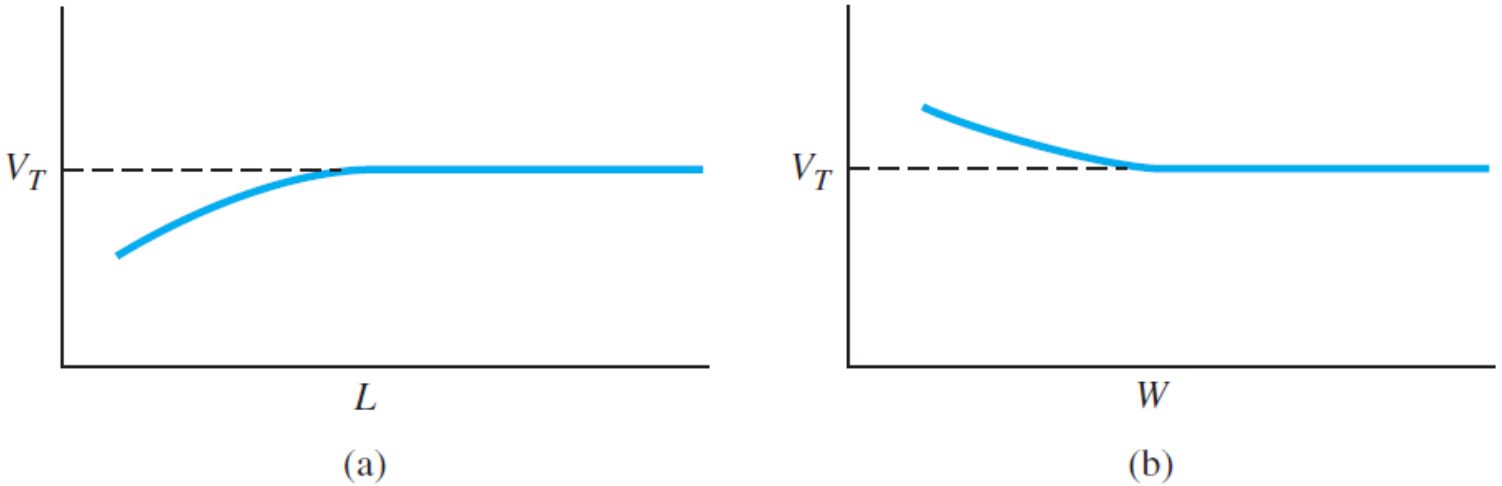
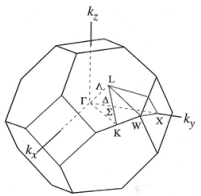
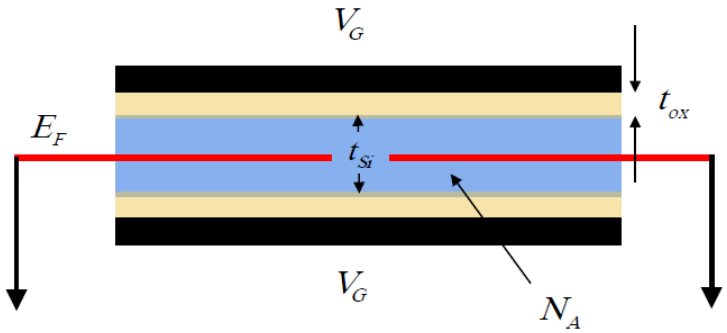


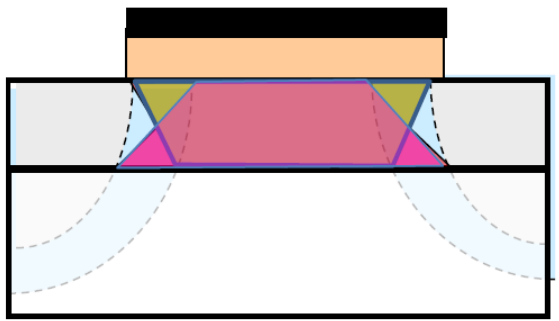
Figure 11.20 | Qualitative variation of threshold voltage (a) with channel length and (b) with channel width.

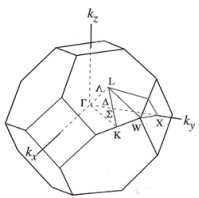


How to Solve This Issue?

- Ultra-thin Body SOI

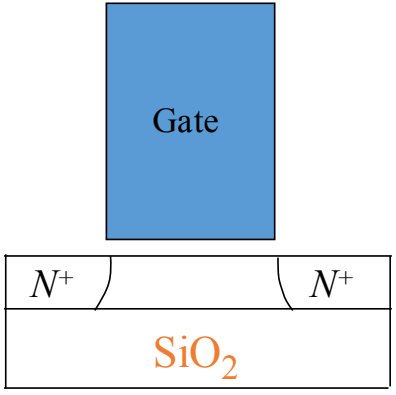




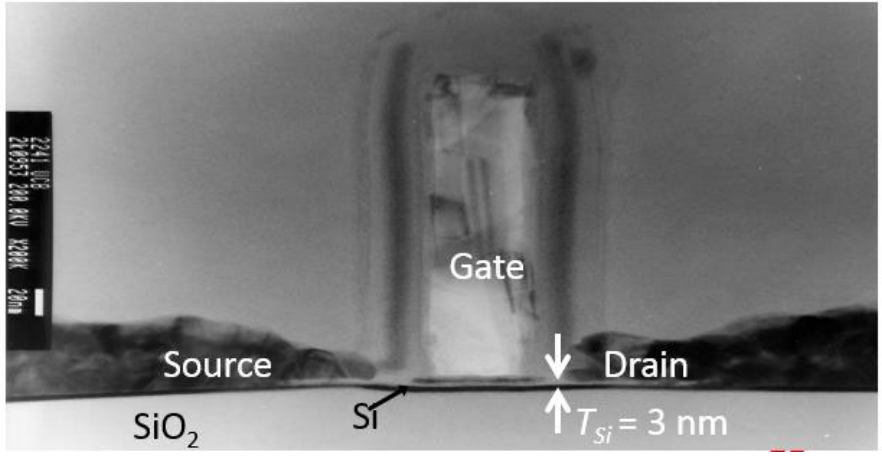


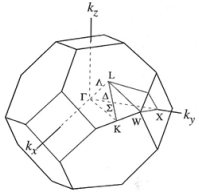
Ultra-Thin-Body MOSFET and SOI

- UTB MOSFET built on ultra thin silicon film on an insulator (SiO_2).
- Since the silicon film is very thin, perhaps less than 10nm, no leakage path is very far from the gate.

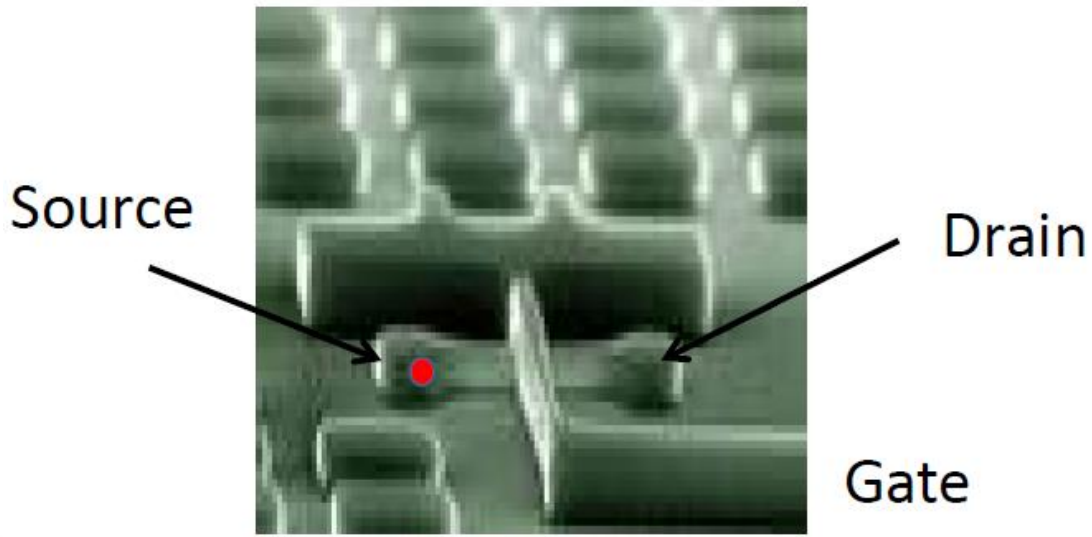


Electron Micrograph of UTB MOSFET

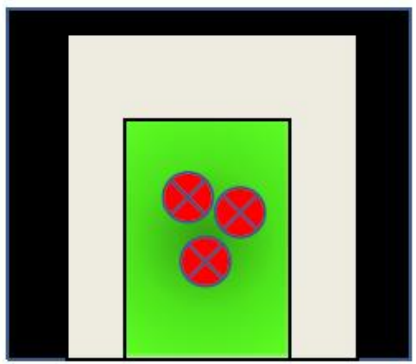


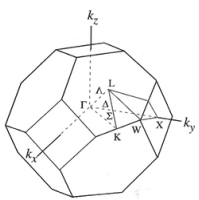


Example: FINFET, OmegaFET, X-FET

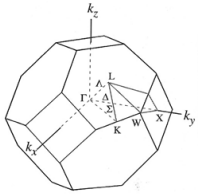


Cross-section

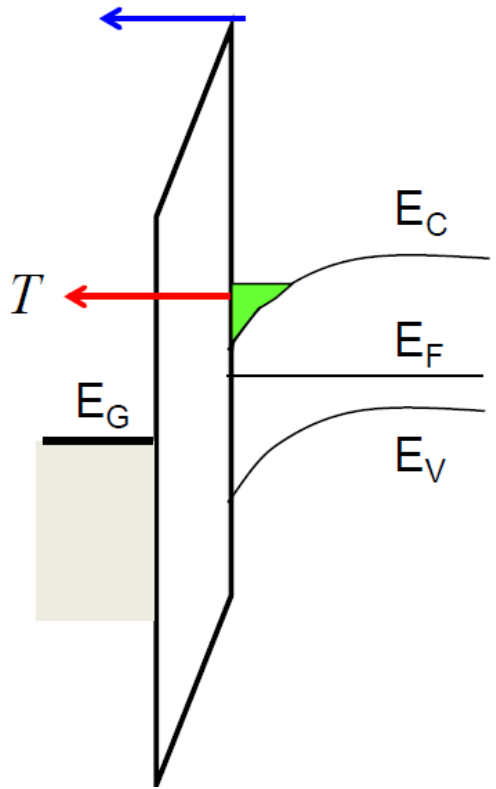


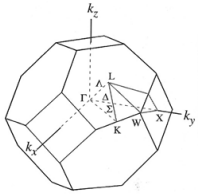


Breakdown of MOSFET



Oxide Breakdown





Avalanche Breakdown

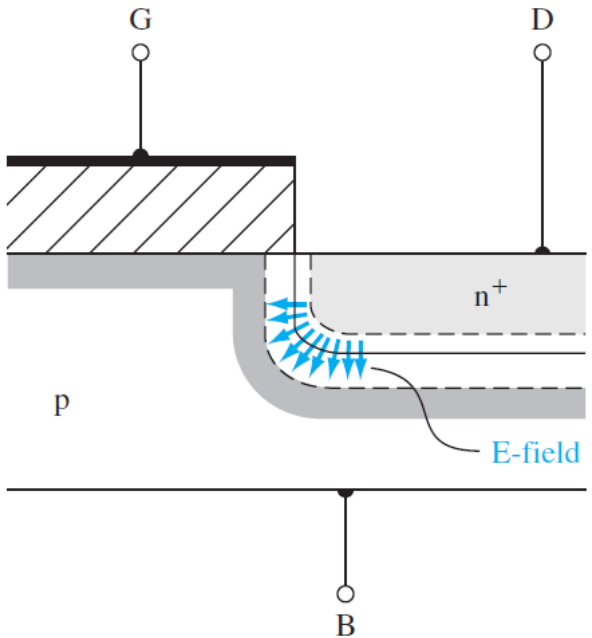
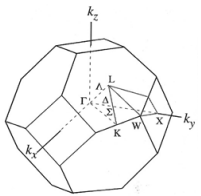


Figure 11.21 | Curvature effect on the electric field in the drain junction.



Near Avalanche and Snapback Breakdown

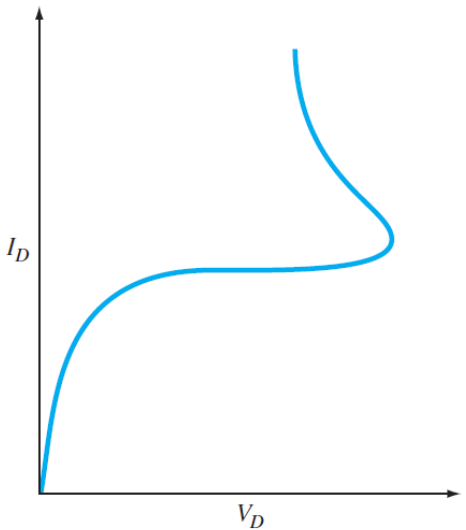


Figure 11.22 | Current–voltage characteristic showing the snapback breakdown effect.

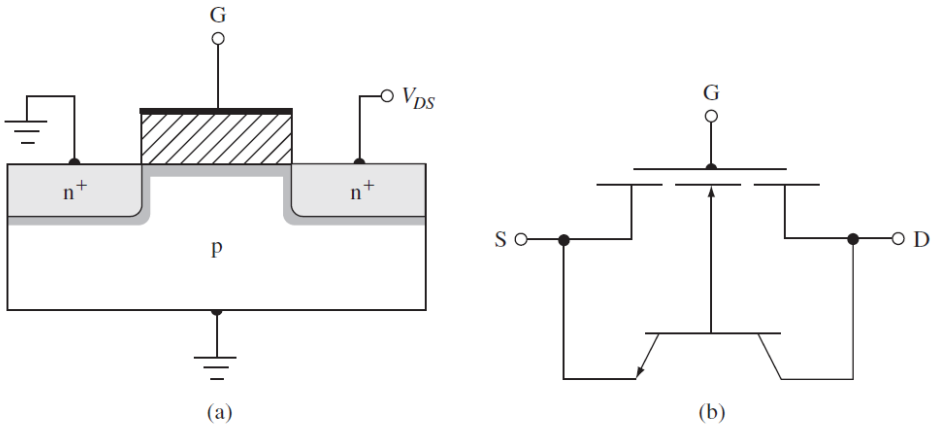
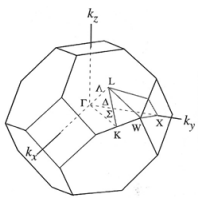


Figure 11.23 | (a) Cross section of the n-channel MOSFET. (b) Equivalent circuit including the parasitic bipolar transistor.



Near Avalanche and Snapback Breakdown

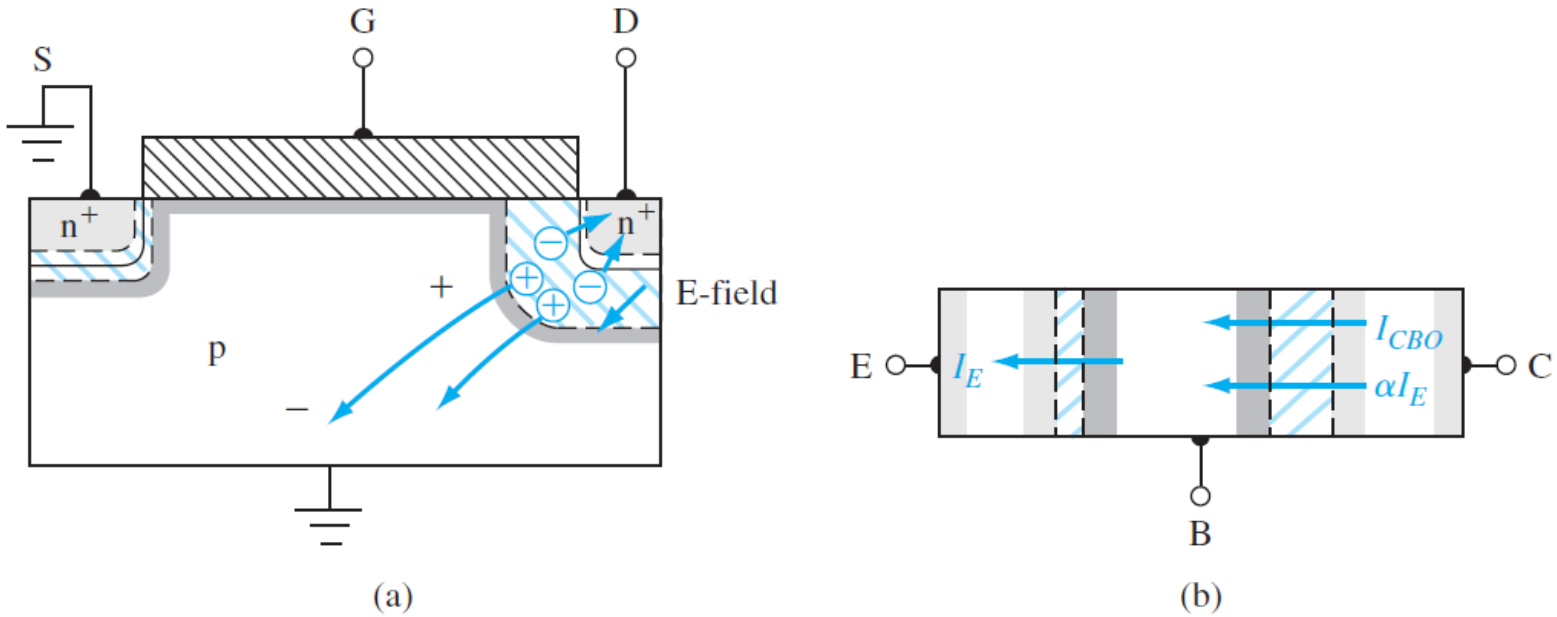
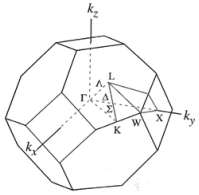


Figure 11.24 | (a) Substrate current–induced voltage drop caused by avalanche multiplication at the drain. (b) Currents in the parasitic bipolar transistor.



Punch-Through Effects

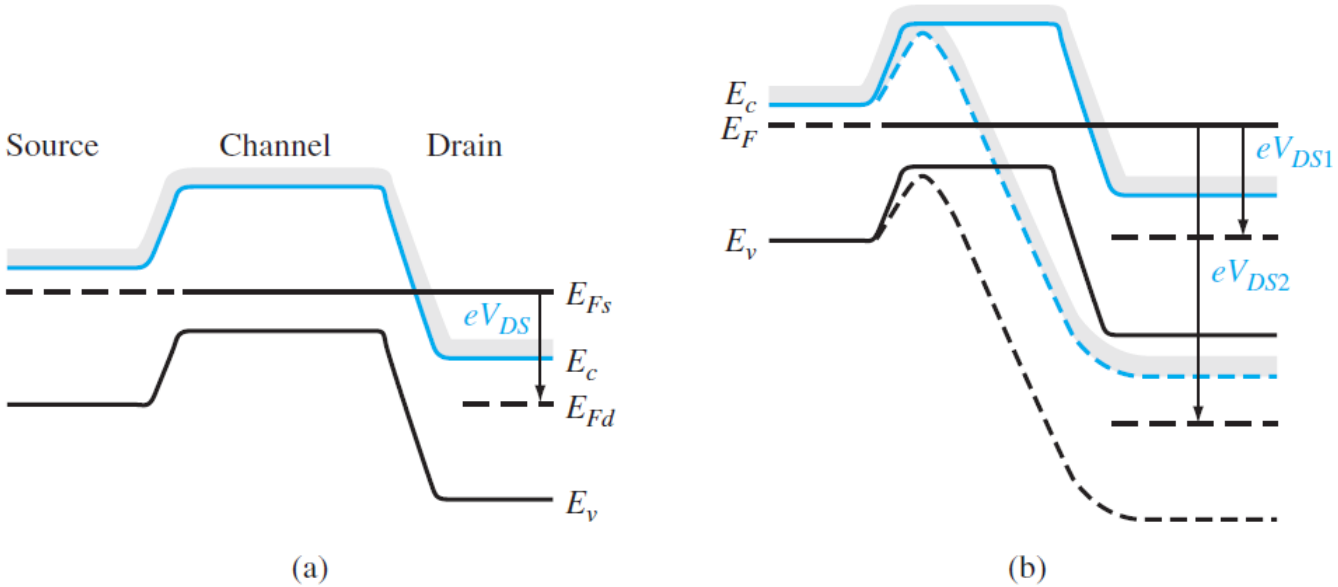
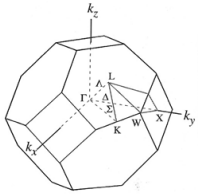


Figure 11.25 | (a) Equipotential plot along the surface of a long-channel MOSFET. (b) Equipotential plot along the surface of a short-channel MOSFET before and after punch-through.



Punch-Through Effects

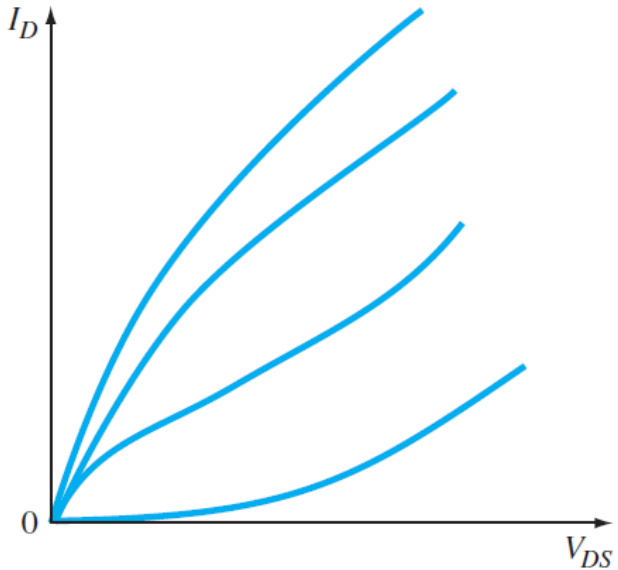
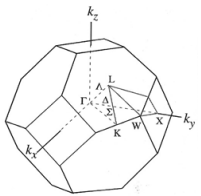
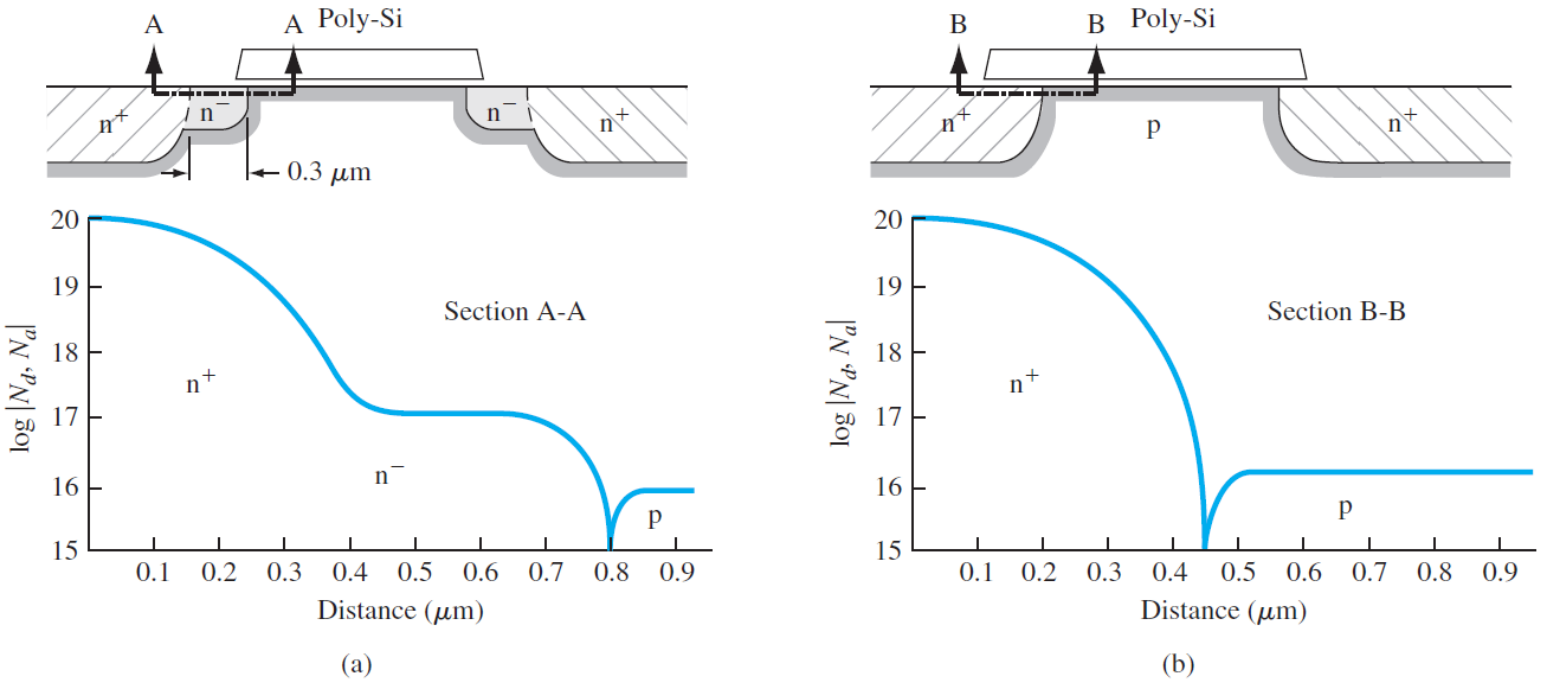


Figure 11.26 | Typical *I–V* characteristics of a MOSFET exhibiting punch-through effects.



Lightly Doped Drain (LDD) Transistor



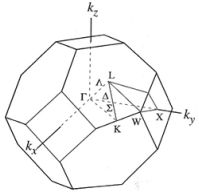


Figure 11.27 | (a) The lightly doped drain (LDD) structure. (b) Conventional structure. (*From Ogura et al. [12].*)

Lightly Doped Drain (LDD) Transistor

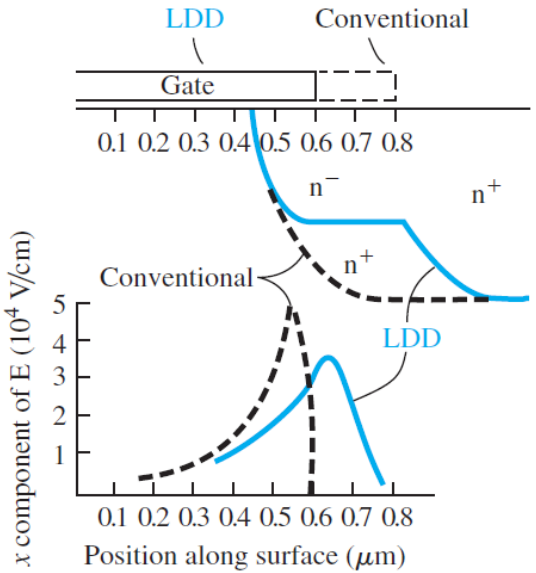


Figure 11.28 | Magnitude of the electric field at the Si–SiO_2 interface as a function of distance; $V_{DS} = 10$ V. $V_{SB} = 2$ V, $V_{GS} = V_T$. (*From Ogura et al. [12].*)

